Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For

Build a Human-AI Workforce with Autonomous AI Agents

TGS Ref No: TGS-2024043854

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version 2.1

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
1.0	12 July 2026	Initial release — 3-day lesson plan for the autonomous-AI-agent labs (Hermes Agent, OpenClaw, Paperclip).	Dr. Alfred Ang
1.1	14 July 2026	Revised to a 2-day plan (Day 1: Topics 1 & 2; Day 2: Topic 3 & final assessment). Hermes expanded to 14 labs, OpenClaw to 12, Paperclip to 8 (34 labs total). Learning outcomes aligned to the Course Proposal / TSC; per-lab reference videos added.	Dr. Alfred Ang
1.2	14 July 2026	Visual redesign: all content slides card-format; YouTube links removed from the deck and Learner Guide (videos remain in the labs); live Hermes screenshots and skill-category grids added; Hermes install lab expanded with the official install methods.	Dr. Alfred Ang
1.3	14 July 2026	Deck restyled in the dark masterclass theme end-to-end; added the Hermes masterclass slides (What is Hermes; The Bet: Compounding Value; platform matrix; Top-10 slash commands; Telegram setup; memory L1-vs-L2 and session-search; what-is-a-tool, built-in tool catalog, tools-in-action, cron scheduling, delegation, kanban orchestration and OpenClaw topic slides); Lab 13 rebuilt as the multi-agent video team (profiles + kanban) with detailed Labs 11-14 guides; fade slide transitions and the live Achievements dashboard screenshot.	Dr. Alfred Ang
1.4	15 July 2026	Security masterclass slides added to the Hermes Security lab (defense-in-depth layers; the gateway's six-check default-deny trust chain; channel lockdown dials; the three approval modes + the Tirith scanner; a live Dangerous-Command approval; YOLO mode; the unrecoverable-command blocklist floor; prompt-injection scanning of context files). Security lab steps expanded with the channel allowlist and approvals.mode.	Dr. Alfred Ang
1.5	15 July 2026	Sixteen concept diagrams imported from the original Mastering OpenClaw masterclass deck into Topic 2 (OpenClaw OS control plane; deployment; hub-and-spoke gateway; tools vs skills; SKILL-on-demand; 6-phase execution loop; exec; HEARTBEAT + autonomic tick; workspace file persistence; context assembly; compaction; defense methods; poisoned-skill anatomy; sessions_spawn; sub-agent routing).	Dr. Alfred Ang
1.6	15 July 2026	Topic 3 (Paperclip) rebuilt as the 'Tertiary AI News Research' company across 6 labs (27-32): install on Windows & Mac; company, CEO & mission; model adaptors; a detailed task backlog whose core task hires the team; the Tavily Search API via Secrets; and hiring the members under the hiring task — with live dashboard screenshots. Natural-language prompt steps added to the Hermes and OpenClaw labs alongside the CLI. Platform logos added to the Three Platforms page. 32 labs total.	Dr. Alfred Ang
1.7	15 July 2026	Sixty-eight content slides migrated from the original 'Mastering OpenClaw' masterclass deck into Topic 2 as framed figures in their matching labs — agent evolution and Agent = Model + Harness, the OpenClaw Wikipedia history and OpenAI acquisition, deployment options and services, model stacks and Mission Control, channels, skills and ClawHub risks, tools (AgentMail, Agent Browser, Firecrawl), system prompt / workspace / context and memory, security lockdown, slash commands, crons, and use cases (NFT certificates, dashboard, cost saving, Google integration).	Dr. Alfred Ang
1.8	15 July 2026	Imported masterclass figures re-blended into the dark deck	Dr. Alfred Ang

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

		theme: original title and copyright strips cropped, white canvases converted to the deck background with text inverted to ivory (embedded screenshots untouched), frameless rendering, and duplicated titles reworded.	
1.9	16 July 2026	Deck curated per trainer review: the bulk masterclass figure imports were removed, keeping nine key figures (Wikipedia history, Agent=Model+Harness, SKILL.md, ClawHub, AgentMail, 10-steps lockdown, HEARTBEAT, autonomic tick, workspace persistence) re-exported at high resolution, with the Topic 2 opening story reordered ahead of Key Concepts. Every CLI step now also shows the natural-language prompt alternative ('Or just ask') across the Hermes and OpenClaw labs, in the deck and Learner Guide.	Dr. Alfred Ang
1.10	16 July 2026	PP assessment updated to v4 — evidence screenshots changed to: Task 1 Hermes interacting on Telegram or the multi-agent Kanban board; Task 2 OpenClaw on Telegram or the TUI; Task 3 the Paperclip setup (company, mission, CEO, hired agents) — with Topic 3 lab references refreshed. A 'RAG vs a Standard LLM Assistant' concept slide added to Core Concepts to ground WA Q2. Final PPT/LP/LG/labs/assessment alignment review.	Dr. Alfred Ang
1.11	16 July 2026	Topic 3 company renamed to 'Altera AI Blogs' with the mission re-pointed at publishing AI-related blogs end to end; the core backlog task is now 'Hire the agents to publish the AI-related blogs end to end' and Lab 32's hiring happens explicitly under it; Lab 29 (Adaptor) and Lab 32 descriptions expanded in full; NEW Lab 33 'Routine & Trigger — Publish Daily at 3pm' (33 labs total), verified with a live 'Run now' pipeline test and two live screenshots.	Dr. Alfred Ang
2.0	18 October 2026	Major revision. Deck redesigned in the 'Agentic AI Applications with Claude Code' house design. Topic 1 rebuilt as Foundations: history of AI, AI agents vs agentic AI, the agentic loop, harness engineering, the brain–perception–action framework, short/long-term memory, world models, single → multi-agent → human–agent collaboration paradigms, agent society, the Oct 2026 agent market (OpenClaw, Hermes Agent, Paperclip, OpenAI Dots, Meta Muse, Block Buzz, Manus Cue and more), Singapore's AgentSea (12,000+ agents) and human–AI use cases in science, maths, drug discovery and business. OpenClaw labs removed; 33 feature labs replaced by 10 connected use-case labs for the fictional Kopi Kaki Co. — Day 1 Hermes Agent (research brief, end-to-end social media campaign, human-in-the-loop publishing, multi-agent video studio, campaign command centre) and Day 2 Paperclip (found the company, hire the team, launch project, weekly video routine, govern & improve) — with mock data, brand assets, skills, instruction PDFs and a local Lab Prompts page. Assessment v5 aligned; FG and AP added.	Dr. Alfred Ang
2.1	29 October 2026	Human–AI landscape rebuilt with practical workflows, research evidence, collaboration modes and humanoid futures; Day 1 single Hermes office automation with gog, Excel, interactive dashboard and professional PPT; Day 2 Kanban and Claude Code video team; SWDA subscription steps and connected GitHub Pages prompts.	Dr. Alfred Ang

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

TABLE OF CONTENTS

Course Information **5**
Learning Outcomes **5**
Topics and Labs **5**
Learning Units (approved Course Proposal TPG-2024060313) **6**
Assessment **6**
Course Schedule **6**
Day 1 — Human–AI landscape and single Hermes agent: office automation, tools, skills and practice (Labs 1–3) 6
Day 2 — Multi-agent teams: Kanban, Claude Code video studio, governance (Labs 4–10), then assessment 7
Lab Reference **7**

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Course Information

Course Title	Build a Human-AI Workforce with Autonomous AI Agents
WSQ Course Reference	TGS-2024043854
Training Provider	Tertiary Infotech Academy Pte Ltd (UEN 201200696W)
Duration	9:30 am–6:30 pm both days. 16 scheduled course hours excluding lunch include 13 hours instruction/practice, 1 hour tea breaks and 2 hours assessment. Tea is included in the scheduled LU allocations; this is not a claim of 14 net instructional hours.
Daily Timing	9:30 am – 6:30 pm (1-hour lunch; tea breaks within training time)
Mode	Instructor-led; hands-on use-case labs on Hermes Agent and Paperclip (learner laptop)
Course scenario	Kopi Kaki Co. — a fictional Singapore kopi chain launching the Gula Melaka Cold Brew (#ShiokColdBrew) — all company data is fictional
Slide deck	Human-AI Workforce with Autonomous AI Agents-v2.1.pptx (218 slides)
Trainer	Dr. Alfred Ang

Learning Outcomes

On completion of this course, learners will be able to:

- LO1: Analyse AI (LLM-based) applications across a range of industries to identify their capabilities and limitations.
- LO2: Establish the relationship between AI algorithm/agent design and chatbot/agent efficiency.
- LO3: Evaluate and improve the effectiveness of AI (RAG and agent) applications in product development.

Topics and Labs

Topic	Content / labs	Slides
Topic 1: Human–AI Collaboration: Landscape, Agents, Use Cases & Futures	Current human–AI interaction landscape; 2023 prompt → 2024 context → 2025 harness → 2026 agents; 48 practical workflows, scientific discovery and humanoid futures; Single agents, teams, orchestration and human control; Agent mechanisms, evidence, permissions and governance. Activity: Human–AI Workforce Canvas.	22–99
Topic 2: Hermes Agent — Single-Agent Office Automation, Tools & Skills	Lab 1 · Install Hermes, Telegram and an evidence brief; Lab 2 · Gmail, Calendar, Drive, Excel, dashboard and PowerPoint; Lab 3 · Office approval and exception handling; Day 2 bridge · Labs 4–5 Kanban and multi-agent work.	100–161
Topic 3: Multi-Agent Teams — Kanban, Claude Code & Governance	Labs 4–5 · Kanban video studio and analysis team; Labs 6–7 · Company and org-chart governance; Lab 8 · Claude Code multi-agent video team; Labs 9–10 · Repeatable work, evaluation and improvement.	162–210

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Learning Units (approved Course Proposal TPG-2024060313)

LU	Learning unit	Delivered through	K & A	Duration
LU1	Foundations of autonomous AI agents and single-agent LLM applications	Topic 1 landscape, canvas and office design clinic + demo + Lab 1A installation and evidence brief	K1, K6, A1, A3	5 h
LU2	Agent and multi-agent design for efficiency, with humans in the loop	Lab 1B + Labs 2–5 tools/skills, human gates, Kanban studio and parallel team; governance introduction	K4, K5, A2, A6	5 h
LU3	Evaluate and improve the effectiveness of an AI-agent workforce in product development	Labs 6–10 company governance, Claude video team, repeatable work and evaluation	K2, K3, A4, A5	4 h
Assessment	WA-SAQ (1 h) + PP (1 h)	Day 2, 4:30–6:30 pm	K1–K6, A1–A6	2 h

Scheduled LU allocations: 5 h + 5 h + 4 h, plus 2 h assessment; these include tea. Net instruction/practice is 13 h. Retained 9:30–18:30 timing was confirmed by the trainer.

Assessment

- Written Assessment (WA) — Short-Answer Questions (SAQ), 1 hour, open book.
- Practical Performance (PP) — three tasks, each with ONE screenshot from a lab (Lab 1 Hermes agent, Lab 4 multi-agent studio, Lab 7 Paperclip org chart) plus written analysis, 1 hour, open book.
- Format: Open Book — course slides, Learner Guide and approved materials only.
- Final assessment is conducted on Day 2 from 4:30 pm.
- A minimum of 75% attendance is required to be eligible for assessment and funding.

Course Schedule

Day 1 — Human–AI landscape and single Hermes agent: office automation, tools, skills and practice (Labs 1–3)

Time	Duration	Topic / Activity	LU	Slides
9:30–9:50	20 min	Welcome, attendance, learning outcomes and official SWDA AI subscription redemption steps	LU1	1–21
9:50–11:05	75 min	**Topic 1 (Part A) — current human–AI landscape; history; agents and agentic AI; practical use-case gallery; science; future humanoid collaboration**	LU1	22–64
11:05–11:20	15 min	Tea break (included in scheduled LU time)	LU1	—
11:20–12:40	80 min	**Topic 1 (Part B) — collaboration modes; single and multi-agent patterns; orchestration; tools, skills, memory, RAG, permissions, governance and evidence**	LU1	65–94
12:40–13:00	20 min	**Activity — design the Kopi Kaki launch-office canvas, inputs, outputs and human decisions**	LU1	95–97
13:00–13:30	30 min	**Office workflow design clinic — analyse task ownership, tool contracts and acceptance tests**	LU1	95–97
13:30–14:30	60 min	Lunch break	—	—
14:30–14:45	15 min	**Trainer demo — OpenAI Dots office workflow, tool evidence and human control; Hermes fallback**	LU1	98–99

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

14:45–15:30	45 min	Lab 1: Activity 1A — install Hermes, connect Telegram and produce the source-linked brief (PP Q1 preparation)	LU1	128–133
15:30–16:00	30 min	Lab 1: Activity 1B — test tools, memory recall and evidence; analyse the agent design	LU2	128–133
16:00–16:15	15 min	Tea break (included in scheduled LU time)	LU2	—
16:15–17:45	90 min	Lab 2: Single-agent launch office — gog Gmail/Calendar/Drive, Excel analysis, interactive dashboard and professional PowerPoint	LU2	134–139
17:45–18:15	30 min	Lab 3: Human approval, office exceptions and safe internal reporting	LU2	140–146
18:15–18:30	15 min	Day 1 recap, Q&A and PM digital attendance	LU2	161

Scheduled time excluding lunch: 480 minutes (8 hours), including tea and any assessment.

Day 2 — Multi-agent teams: Kanban, Claude Code video studio, governance (Labs 4–10), then assessment

Time	Duration	Topic / Activity	LU	Slides
9:30–9:45	15 min	Day 1 recap and AM attendance; prepare the multi-agent board	LU2	161
9:45–10:35	50 min	Lab 4: Multi-Agent Video Studio — Kanban roles, dependencies, production, audit and human release gate	LU2	147–153
10:35–11:10	35 min	Lab 5: Multi-Agent Campaign Command Centre — parallel analysis, verifier and synthesis	LU2	154–160
11:10–11:25	15 min	Tea break (included in scheduled LU time)	LU2	—
11:25–11:30	5 min	**Topic 3 — company-level governance and team accountability**	LU2	162–166
11:30–12:15	45 min	Lab 6: Found Kopi Kaki Growth Studio — mission, company and governed CEO	LU3	167–173
12:15–13:15	60 min	Lunch break	—	—
13:15–13:50	35 min	Lab 7: Hire the Marketing Team through the Board gate — org-chart evidence for PP Q3	LU3	174–191
13:50–14:40	50 min	Lab 8: Claude Code Agent Team — create and audit a launch video with independent teammates	LU3	192–196
14:40–15:10	30 min	Lab 9: Repeatable Weekly Video Pipeline — routine, verification and Board review	LU3	197–202
15:10–15:25	15 min	Tea break (included in scheduled LU time)	LU3	—
15:25–16:00	35 min	Lab 10: Govern, Measure and Improve — compare single-agent and team workflows	LU3	203–209
16:00–16:15	15 min	Course recap, practice questions and Q&A	LU3	211–214
16:15–16:30	15 min	**Briefing for Assessment**	LU3	17
16:30–17:30	60 min	**Written Assessment (WA) — 1 hour, open book**	Assessment	215–218
17:30–18:30	60 min	**Practical Performance (PP) — 1 hour, open book; TRAQOM survey and PM / assessment attendance**	Assessment	215–218

Scheduled time excluding lunch: 480 minutes (8 hours), including tea and any assessment.

Lab Reference

Lab	Title	Paradigm	LO	Slides
Lab 1 (Day 1)	Activity 1 — Install Hermes, Connect Telegram & Build the Launch Brief	Single agent	LO1	128–133
Lab 2 (Day 1)	Single-Agent Launch Office — Google, Excel, Dashboard & Professional PPT	Single agent + four skills	LO1	134–139
Lab 3 (Day 1)	Human-in-the-Loop Office Approval & Exception Handling	Human–agent collaboration	LO2	140–146
Lab 4 (Day	Multi-Agent Video Studio — Kanban, Production &	Multi-agent (ordered	LO2	147–153

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

2)	Human Review	pipeline)		
Lab 5 (Day 2)	Multi-Agent Campaign Command Centre — Measure, Optimise & Report	Multi-agent (parallel swarm)	LO2	154–160
Lab 6 (Day 2)	Found Kopi Kaki Growth Studio — an AI Company You Govern	Human-governed company of agents	LO2	167–173
Lab 7 (Day 2)	Hire the Marketing Team Through the Board Approval Gate	Multi-agent organisation	LO3	174–191
Lab 8 (Day 2)	Claude Code Agent Team — Produce the Launch Video	Claude team + human gate	LO3	192–196
Lab 9 (Day 2)	Automate the Weekly Video Pipeline with a Routine and a Board Gate	Scheduled multi-agent workflow	LO3	197–202
Lab 10 (Day 2)	Govern, Measure & Improve Your Human–AI Workforce	Workforce governance	LO3	203–209

Lab materials: labs/labNN-*/ (README, LAB-NN-Instructions.pdf, COPY-PROMPTS.txt, data/) and the Lab Prompts page prompts/index.html in the course repository.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg